

1	Course Name:
	Artificial Intelligence and Pattern Recognition
2	Course Code:
	IRG61004
3	Year and Semester Offered:
	Year 1 Semester 2
4	Credit Hours Value:
	4 Credit Hours [4 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis:
	This course introduces students to the field of artificial intelligence and pattern recognition. The high level focus will be on knowledge, reasoning, supervised and unsupervised learning, statistical pattern recognition, syntactic (structural) pattern recognition, and neural pattern recognition. Students will also writing technical paper on AI and Pattern Recognition.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Evaluates various types of AI and pattern recognition algorithms. 2. Constructs algorithm based on pattern recognition techniques in problem solving. 3. Defends technical paper with understanding of managerial and business practices competencies in AI and Pattern Recognition.
8	Content Outline:
	Topic
	1. Introduction to Artificial Intelligence (AI)
	2. Introduction and General Pattern Recognition Concerns
	3. Statistical Pattern Recognition
	4. Syntactic (Structural) Pattern Recognition
	5. Neural Pattern Recognition
	6. Writing technical paper in Artificial Intelligence and Pattern Recognition
9	References:
	1. Christopher M. Bishop. Pattern Recognition and Machine Learning (Revised Edition). Springer-Verlag, 2016, ISBN-10: 1493938436 2. Richard O. Duda, Peter E. Hart, David G. Stork, Pattern Classification, Wiley, 2001, ISBN-10: 04710566933.